

# Python Virtual Environment

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## Python Virtual Environment

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If you are using the factory image we provided, manual installation and configuration are not required.

In earlier versions of the operating system, you could install programs using Python's pip tool:

```
pip install buildhat
```

However, in newer versions of the Raspberry Pi OS system, this is not allowed. If you try to install Python packages system-wide, you will receive an error similar to the following:

```
error: externally-managed-environment

× This environment is externally managed
➤ To install Python packages system-wide, try apt install
  python3-xyz, where xyz is the package you are trying to
  install.

  If you wish to install a non-Debian-packaged Python package,
  create a virtual environment using python3 -m venv path/to/venv.
  Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
  sure you have python3-full installed.

  For more information visit http://rptl.io/venv

note: If you believe this is a mistake, please contact your Python installation or OS distribution
provider. You can override this, at the risk of breaking your Python installation or OS, by passing
--break-system-packages.
hint: See PEP 668 for the detailed specification.
```

In the Jupyter Lab environment setup, we used another method to eliminate this warning. However, here we only introduce how to set up a Python virtual environment.

## Python Virtual Environment

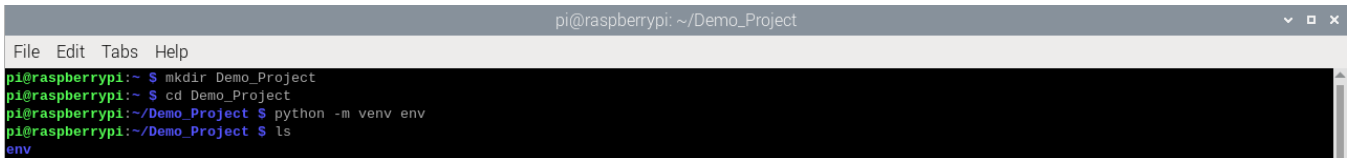
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To use a virtual environment, you need to create a container to store the environment.

### Create Virtual Environment

Create and enter the Demo\_Project folder, then use Python's built-in module venv to create a virtual environment named env:

```
mkdir Demo_Project
cd Demo_Project
python -m venv env
```

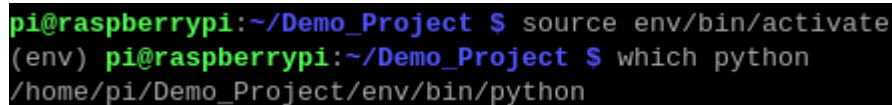
A screenshot of a terminal window titled 'pi@raspberrypi: ~/Demo\_Project'. The terminal shows the following commands and output: 'pi@raspberrypi:~\$ mkdir Demo\_Project', 'pi@raspberrypi:~\$ cd Demo\_Project', 'pi@raspberrypi:~/Demo\_Project\$ python -m venv env', and 'pi@raspberrypi:~/Demo\_Project\$ ls'. The output of the last command is 'env'.

After successful operation, an env folder will be added under the folder.

## Activate Virtual Environment

This directory contains a complete Python distribution. Activate the virtual environment and make this Python version the currently used version:

```
source env/bin/activate
```

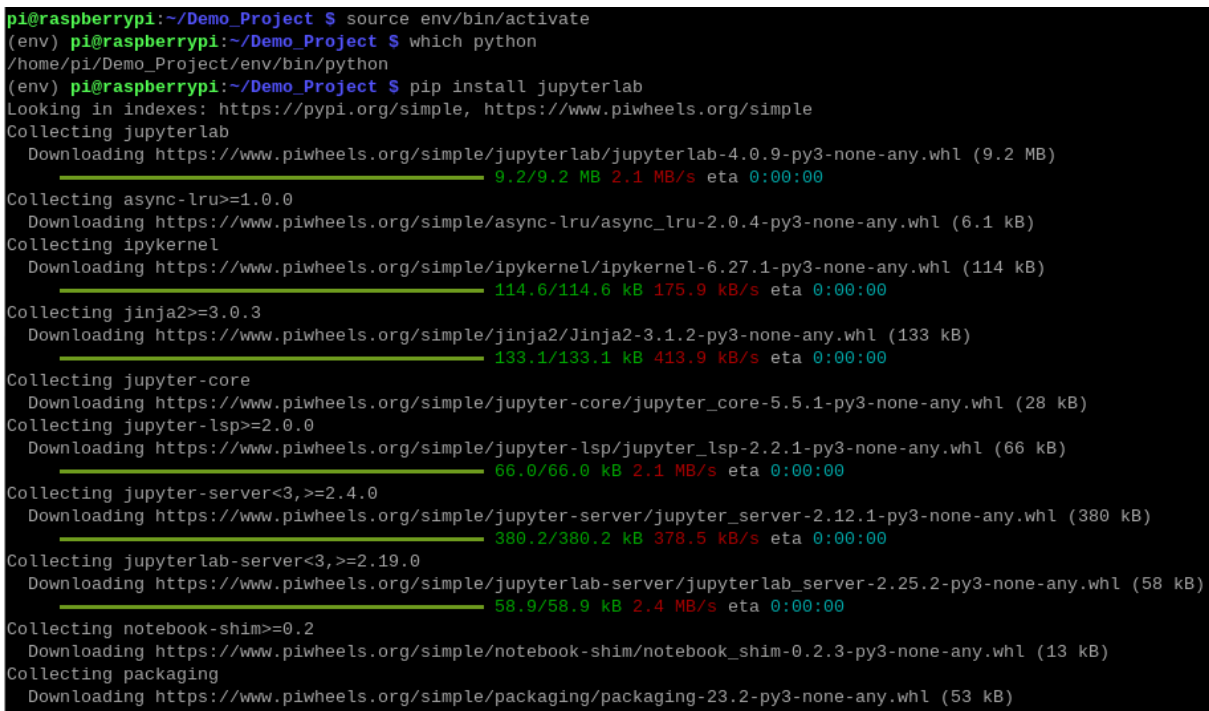
A screenshot of a terminal window showing the activation of the virtual environment. The commands and output are: 'pi@raspberrypi:~/Demo\_Project\$ source env/bin/activate', '(env) pi@raspberrypi:~/Demo\_Project\$ which python', and '/home/pi/Demo\_Project/env/bin/python'.

At this point, we will no longer use the system Python, but instead use the Python version contained in the virtual environment:

- Any changes made here will not affect your system Python;
- Any new modules installed in the environment will not affect the system Python.

## Install Third-Party Packages

```
pip install jupyterlab
```

A screenshot of a terminal window showing the installation of jupyterlab and its dependencies. The commands and output are: 'pi@raspberrypi:~/Demo\_Project\$ source env/bin/activate', '(env) pi@raspberrypi:~/Demo\_Project\$ which python', '/home/pi/Demo\_Project/env/bin/python', '(env) pi@raspberrypi:~/Demo\_Project\$ pip install jupyterlab', 'Looking in indexes: https://pypi.org/simple, https://www.piwheels.org/simple', 'Collecting jupyterlab', 'Downloading https://www.piwheels.org/simple/jupyterlab/jupyterlab-4.0.9-py3-none-any.whl (9.2 MB)', '9.2/9.2 MB 2.1 MB/s eta 0:00:00', 'Collecting async-lru>=1.0.0', 'Downloading https://www.piwheels.org/simple/async-lru/async\_lru-2.0.4-py3-none-any.whl (6.1 kB)', 'Collecting ipykernel', 'Downloading https://www.piwheels.org/simple/ipykernel/ipykernel-6.27.1-py3-none-any.whl (114 kB)', '114.6/114.6 kB 175.9 kB/s eta 0:00:00', 'Collecting Jinja2>=3.0.3', 'Downloading https://www.piwheels.org/simple/jinja2/Jinja2-3.1.2-py3-none-any.whl (133 kB)', '133.1/133.1 kB 413.9 kB/s eta 0:00:00', 'Collecting jupyter-core', 'Downloading https://www.piwheels.org/simple/jupyter-core/jupyter\_core-5.5.1-py3-none-any.whl (28 kB)', 'Collecting jupyter-lsp>=2.0.0', 'Downloading https://www.piwheels.org/simple/jupyter-lsp/jupyter\_lsp-2.2.1-py3-none-any.whl (66 kB)', '66.0/66.0 kB 2.1 MB/s eta 0:00:00', 'Collecting jupyter-server<3,>=2.4.0', 'Downloading https://www.piwheels.org/simple/jupyter-server/jupyter\_server-2.12.1-py3-none-any.whl (380 kB)', '380.2/380.2 kB 378.5 kB/s eta 0:00:00', 'Collecting jupyterlab-server<3,>=2.19.0', 'Downloading https://www.piwheels.org/simple/jupyterlab-server/jupyterlab\_server-2.25.2-py3-none-any.whl (58 kB)', '58.9/58.9 kB 2.4 MB/s eta 0:00:00', 'Collecting notebook-shim>=0.2', 'Downloading https://www.piwheels.org/simple/notebook-shim/notebook\_shim-0.2.3-py3-none-any.whl (13 kB)', 'Collecting packaging', 'Downloading https://www.piwheels.org/simple/packaging/packaging-23.2-py3-none-any.whl (53 kB)', '53.0/53.0 kB 133.0 kB/s eta 0:00:00'.

View modules installed in the current Python version:

```
pip list
```

```
(env) pi@raspberrypi:~/Demo_Project $ pip list
Package                               Version
-----
anyio                                 4.2.0
argon2-cffi                          23.1.0
argon2-cffi-bindings                 21.2.0
arrow                                 1.3.0
asttokens                             2.4.1
asvnc-lru                             2.0.4
jupyter_client                        8.6.0
jupyter_core                          5.5.1
jupyter-events                        0.9.0
jupyter-lsp                           2.2.1
jupyter_server                       2.12.1
jupyter_server_terminals              0.5.0
jupyterlab                            4.0.9
jupyterlab_pygments                   0.3.0
jupyterlab_server                     2.25.2
```

## Exit Virtual Environment

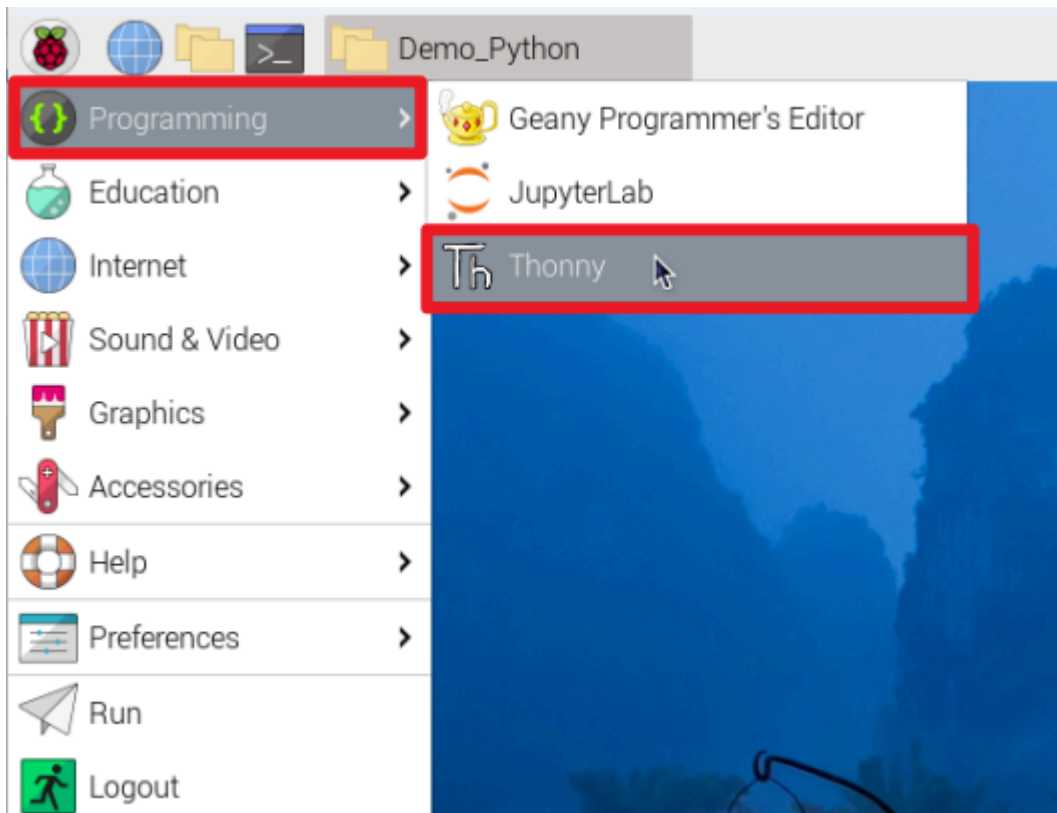
```
deactivate
```

```
(env) pi@raspberrypi:~/Demo_Project $ deactivate
pi@raspberrypi:~/Demo_Project $
```

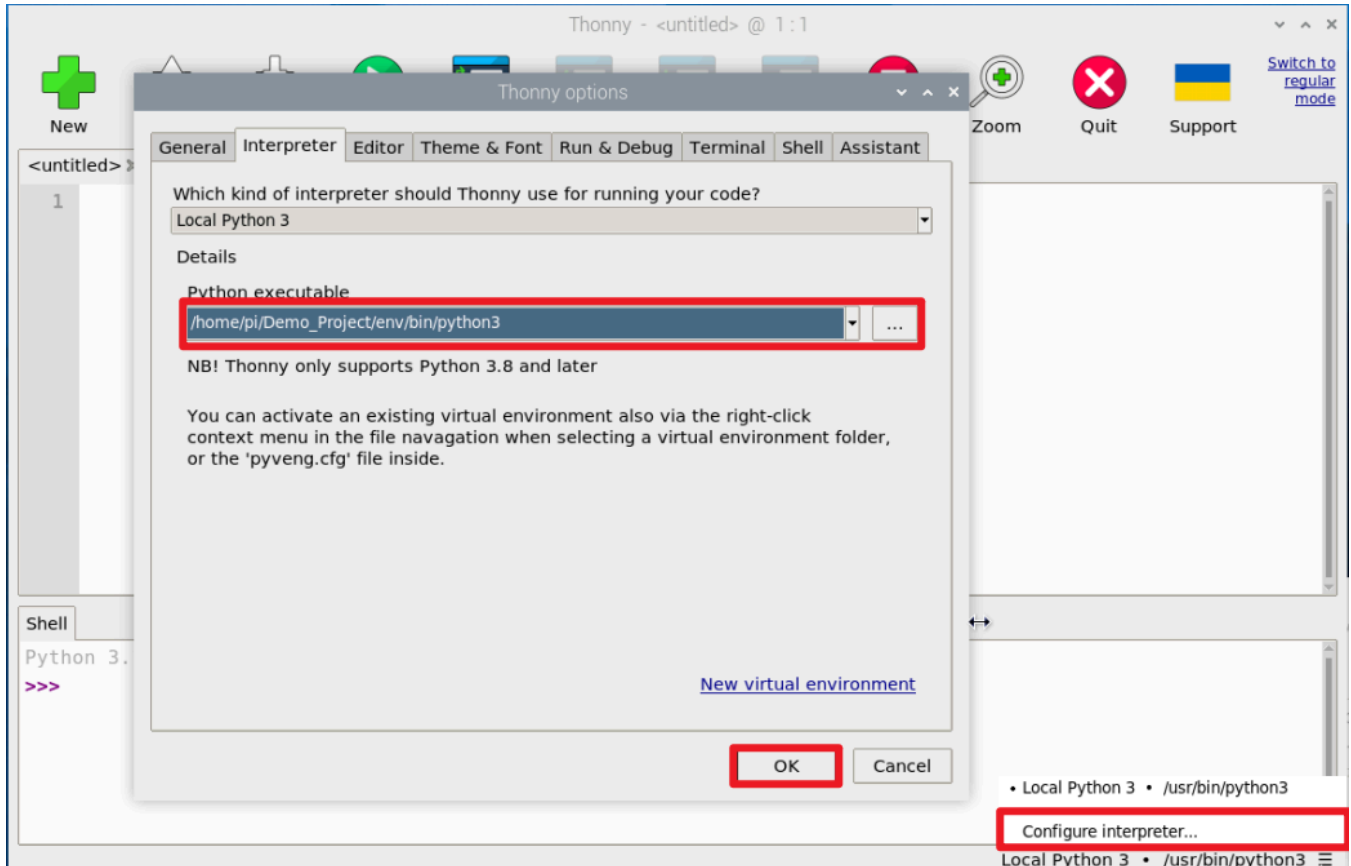
## Thonny Editor

When using Python on Raspberry Pi, Thonny is the system's default editor, and the interpreter uses system Python.

## Switch Virtual Environment Python Version



Click the interpreter menu in the lower right corner of the Thonny interface to switch to the Python virtual environment: Add according to the virtual environment path.



Configuration successful:

